



INDIAN INSTITUTE OF TECHNOLOGY, JODHPUR
Department of Information Technology

AI Job Replacement & Workforce Transformation Analysis

Visualization & Storytelling with Data:
CASE STUDY REPORT

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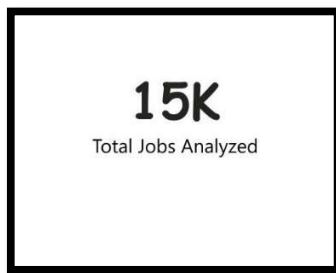
College: IIT Jodhpur

Introduction

Artificial Intelligence is rapidly transforming industries, job roles, and workforce requirements across the world. This case study analyzes the impact of AI automation on jobs, salaries, workforce disruption, and future job growth using Power BI visualizations.

The objective of this analysis is to identify industries and job categories most vulnerable to automation, understand future workforce trends, and provide strategic recommendations for employees and students in an AI-driven economy.

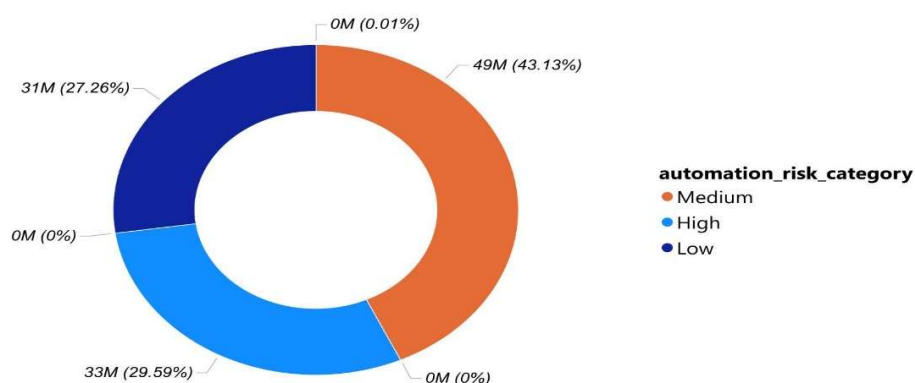
Q1. Total Jobs Analyzed



The dataset contains a large global workforce sample covering multiple industries, countries, and job categories. The KPI card highlights the total number of jobs analyzed in the study.

Q2. High Risk vs Medium Risk vs Low Risk Jobs

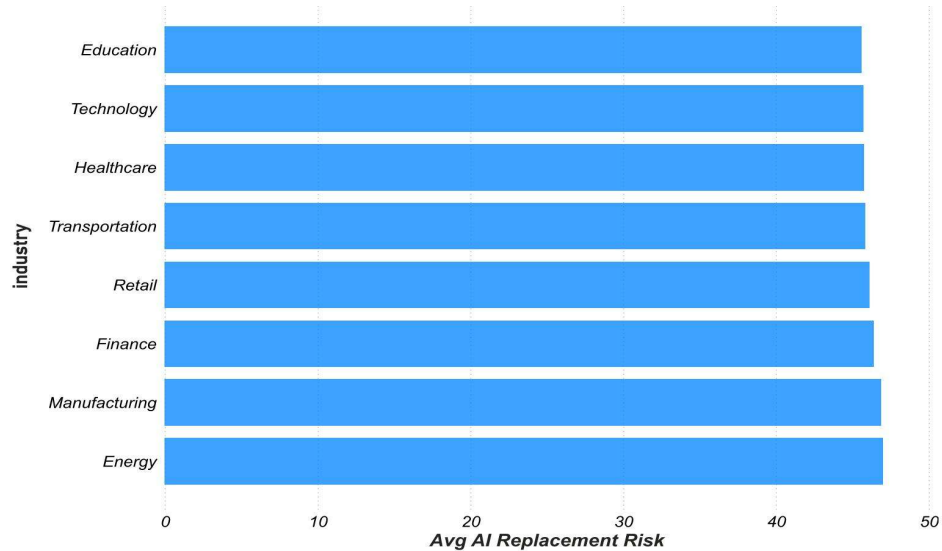
AI Automation Risk Distribution



The analysis shows that a significant portion of jobs fall under Medium and High automation risk categories, indicating that AI is expected to impact a large part of the workforce in the coming years.

Q3. Industry with Highest AI Replacement Risk

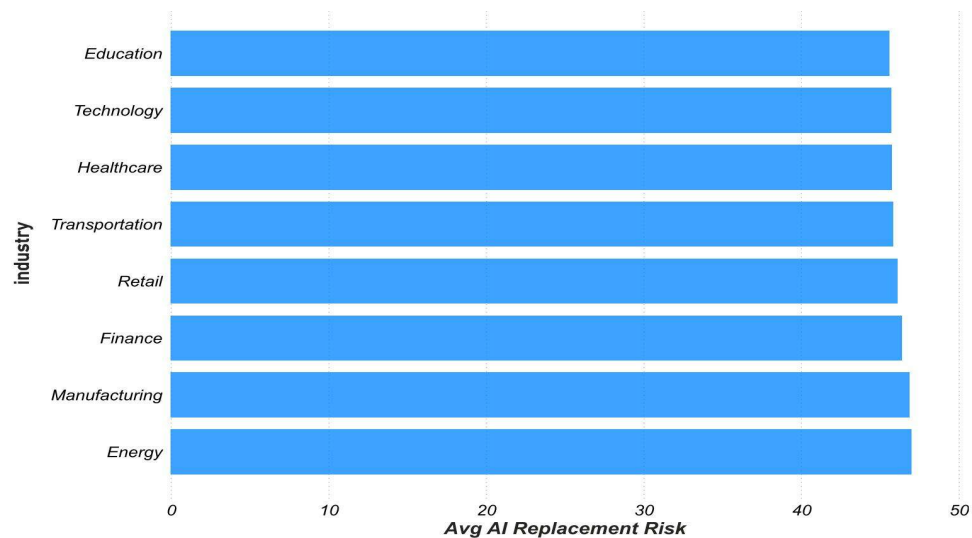
Average AI Replacement Risk by Industry



The dashboard indicates that industries involving repetitive and process-driven tasks demonstrate the highest AI replacement risk due to increased automation capability.

Q4. Safest Industry from AI Automation

Industries Safest from AI Automation



Industries requiring creativity, human interaction, strategic thinking, and specialized expertise appear relatively safer from AI automation.

Q5. KPI Dashboard Insights

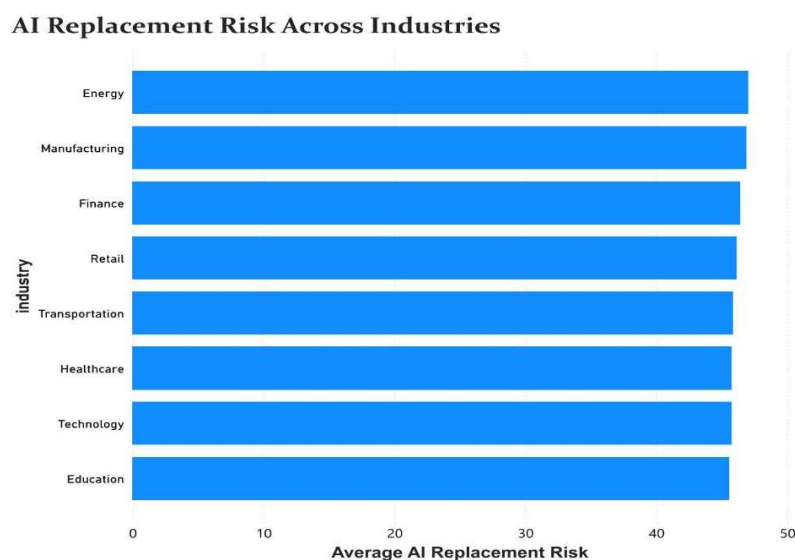


The KPI section provides a quick summary of:

- Total jobs analyzed
- Average automation risk
- Most affected industry
- Safest industry

This allows management to quickly understand the overall AI disruption landscape.

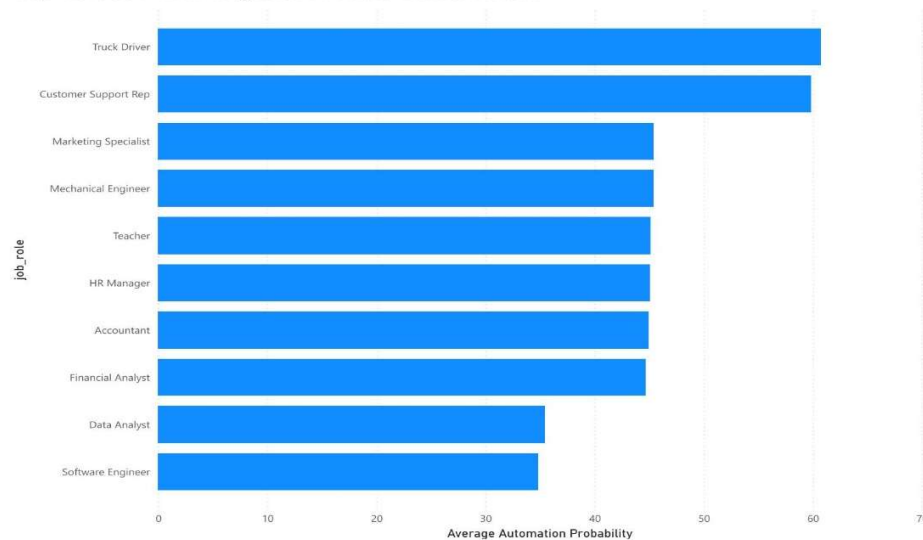
Q6. AI Replacement Risk Across Industries



The bar chart comparison shows that automation risk differs significantly across industries, with some sectors being more vulnerable because of repetitive operational tasks.

Q7. Top 10 Jobs with Highest Automation Probability

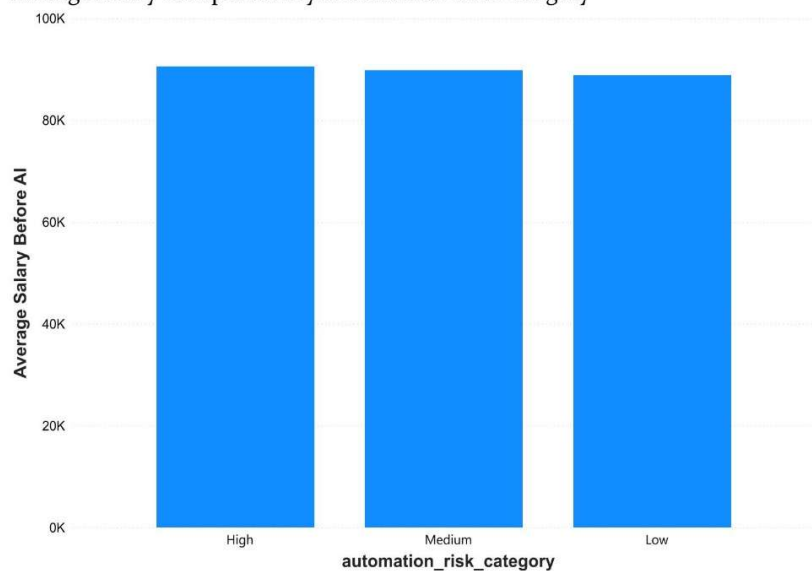
Top 10 Jobs with Highest AI Automation Risk



The visualization identifies the most automation-prone jobs, primarily involving repetitive, predictable, and rule-based activities.

Q8. High-Risk vs Low-Risk Salary Comparison

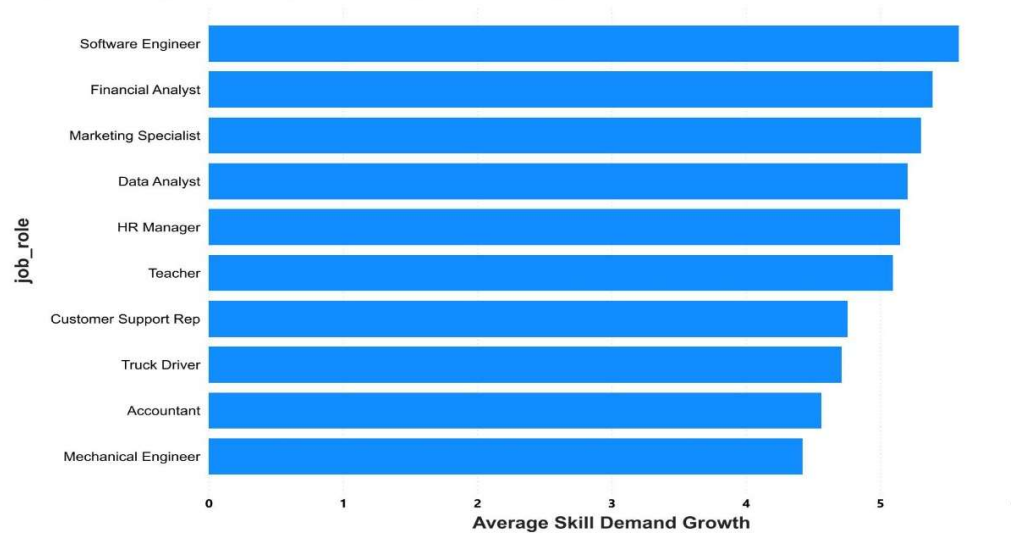
Average Salary Comparison by Automation Risk Category



The salary comparison reveals that high-risk jobs generally have lower salary levels, while low-risk jobs tend to involve specialized and higher-paying roles.

Q9. Job Categories Expected to Grow Despite AI

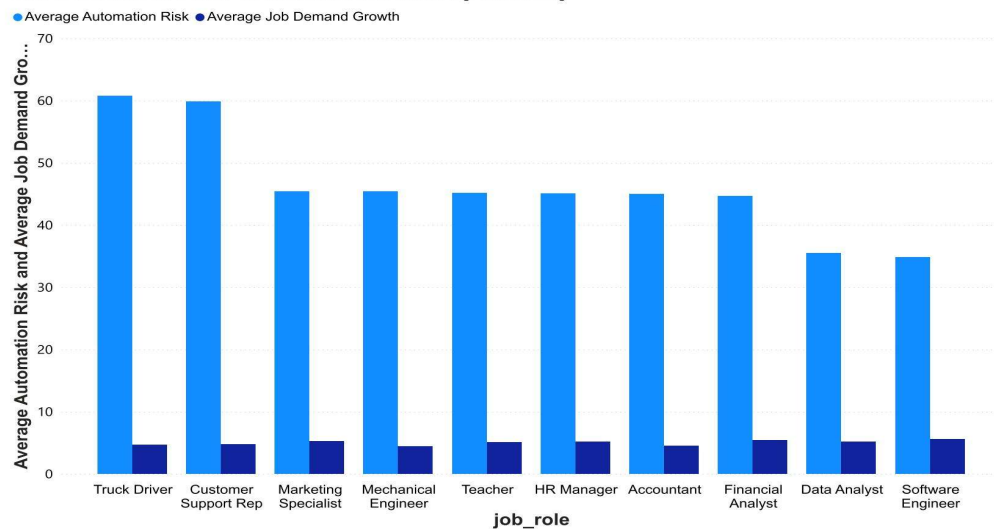
Top Growing Job Categories Despite AI Adoption



The analysis shows that technical, analytical, and AI-related job categories continue to experience positive future demand growth despite increasing automation.

Q10. Job Demand vs Automation Risk

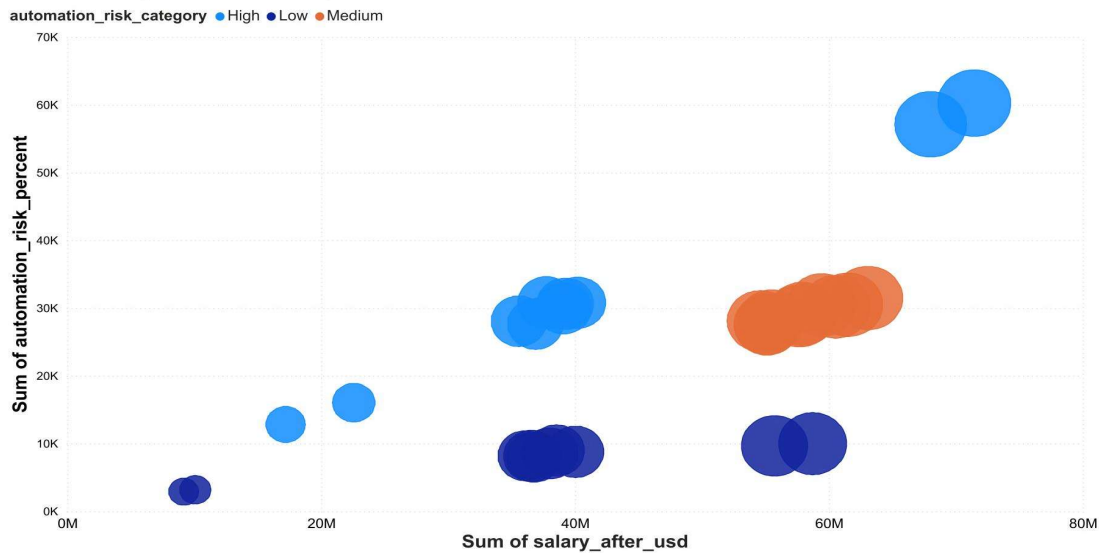
Job Demand Growth vs Automation Risk by Industry



The clustered chart highlights that some industries face both rising automation risk and increasing job demand, suggesting workforce transformation rather than complete job elimination.

Q11. Salary vs Automation Risk

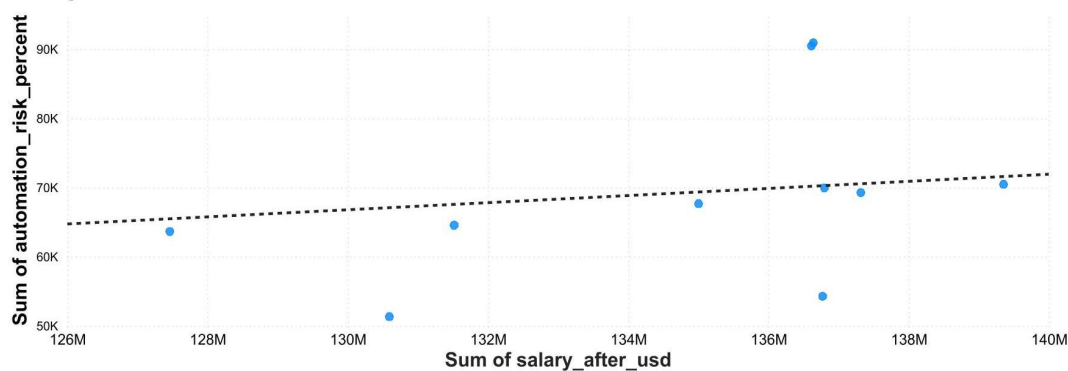
Relationship Between Salary and Automation Risk



The scatter plot indicates that higher-paying jobs generally exhibit lower automation risk, suggesting that specialized and high-skill roles remain more resilient to AI disruption. However, some repetitive technical jobs still face notable automation exposure.

Q12. Are High-Paying Jobs Safer?

Salary vs AI Automation Risk



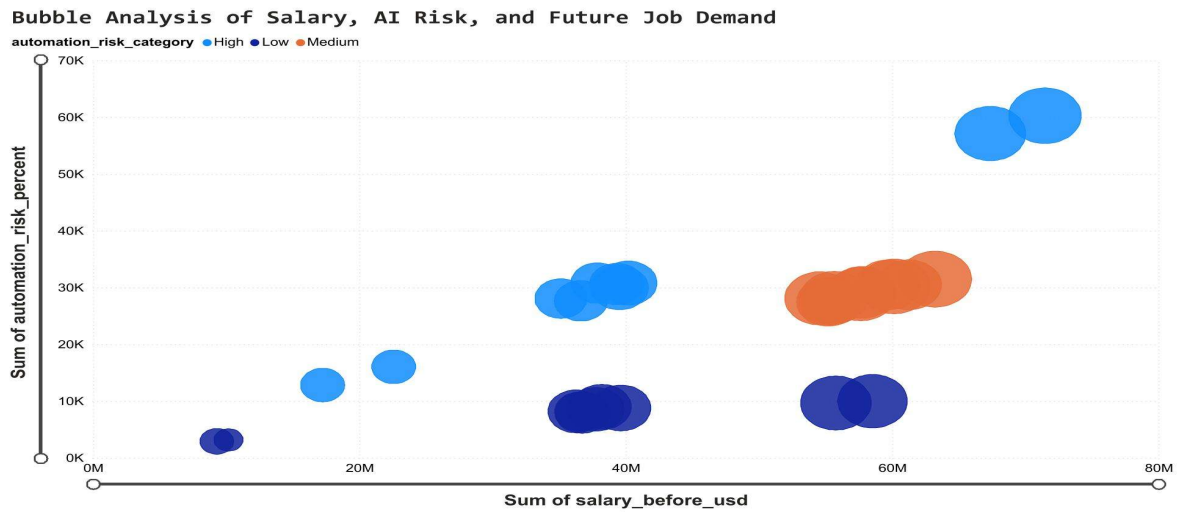
The scatter plot indicates that higher-paying jobs generally exhibit lower automation risk, suggesting that specialized and high-skill roles remain more resilient to AI disruption.

Yes, high-paying jobs appear relatively safer from AI replacement.

Not all high-paying jobs are safe from AI replacement. Some highly paid but repetitive technical roles also show significant automation risk.

The dashboard suggests that higher-paying jobs are relatively safer from AI replacement because they often require advanced decision-making, creativity, and technical expertise.

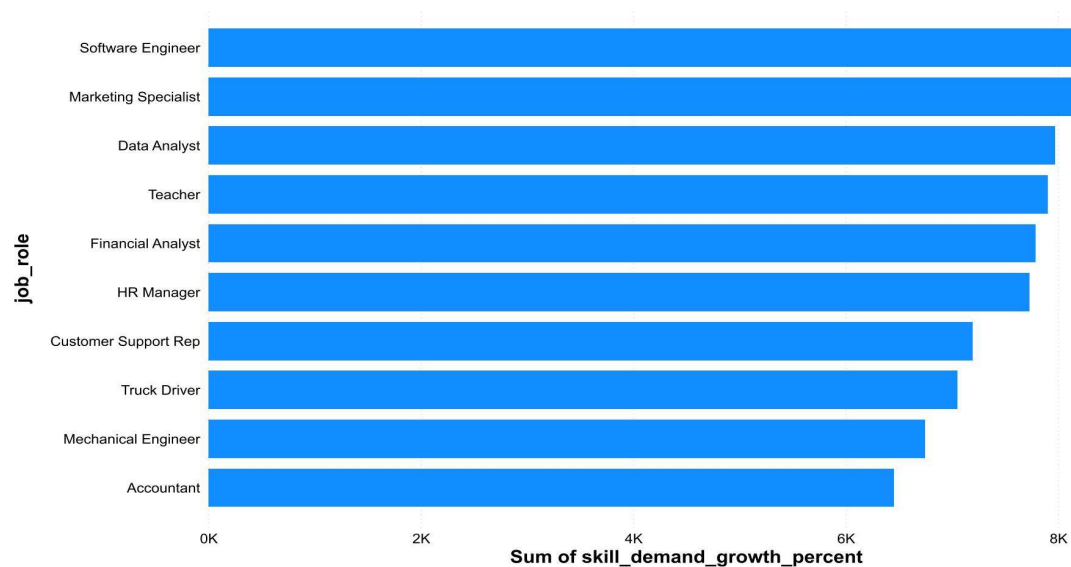
Q13. Bubble Analysis of Salary, Risk, and Job Demand



The bubble chart reveals the relationship between salary, automation risk, and future job demand, helping identify jobs that combine high salary, lower risk, and strong future growth potential.

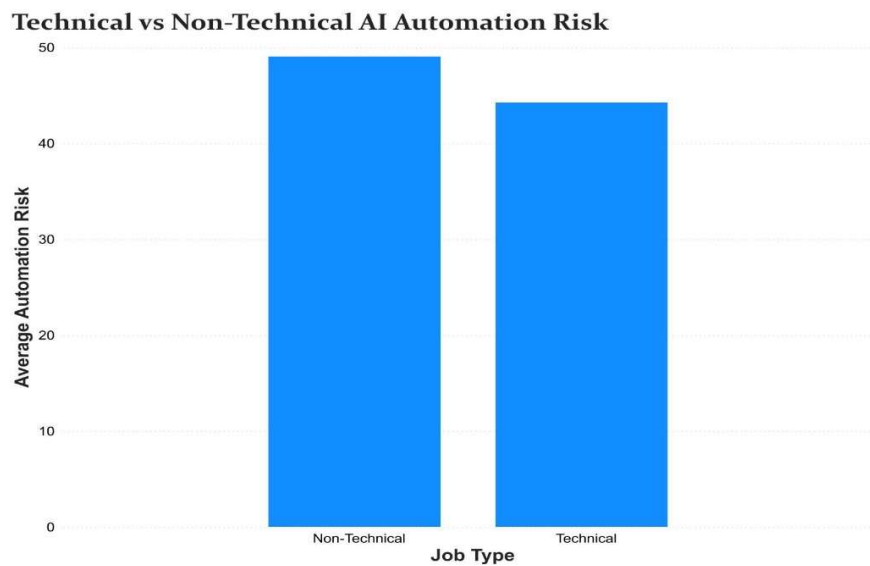
Q14. Valuable Skills in AI-Resistant Jobs

Most Valuable Skills in AI-Resistant Jobs



The visualization shows that AI-resistant jobs are strongly associated with technical expertise, analytical thinking, and higher education requirements. Roles such as Software Engineer and Data Analyst demonstrate stronger future demand growth.

Q15. Technical vs Non-Technical Job Risk



The comparison indicates that non-technical jobs generally face higher automation risk, while specialized technical roles remain comparatively safer from AI disruption.

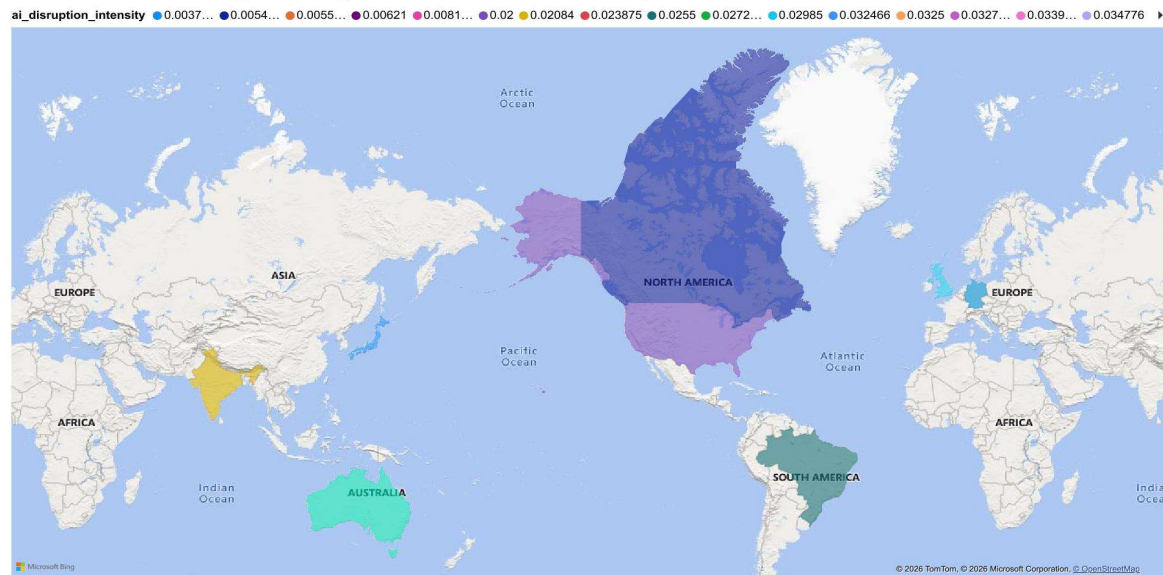
Q16. Global AI Adoption by Country



The geographic map highlights that AI adoption levels vary across countries, indicating differences in technological advancement and automation readiness worldwide.

Q17. Countries with Highest Workforce Disruption

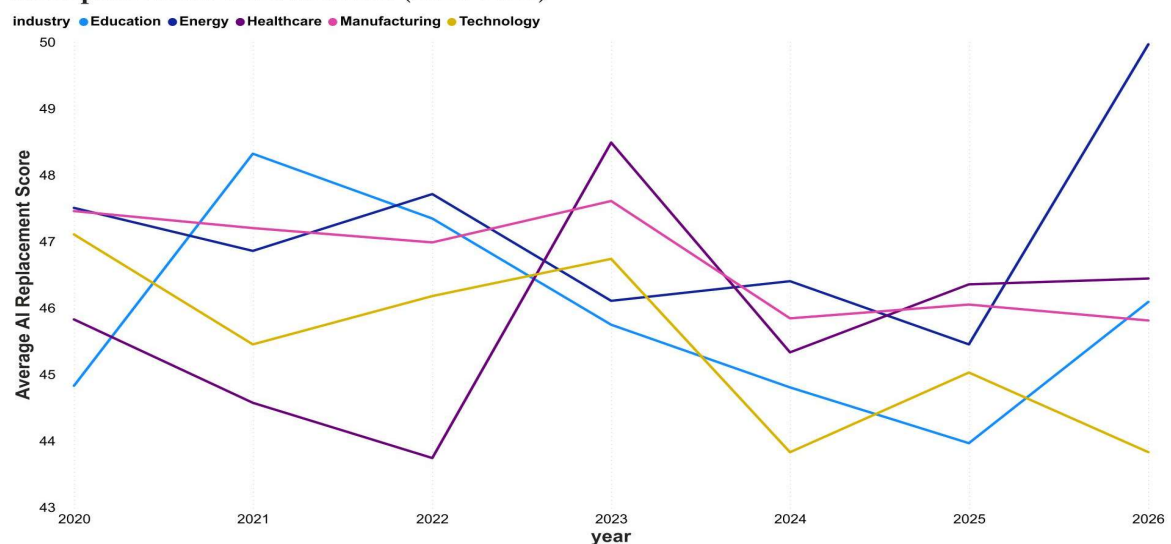
Countries with Highest Expected Workforce Disruption



The workforce disruption analysis identifies regions expected to face major workforce transformation due to rising AI adoption and automation intensity.

Q18. AI Job Replacement Trend Over Time

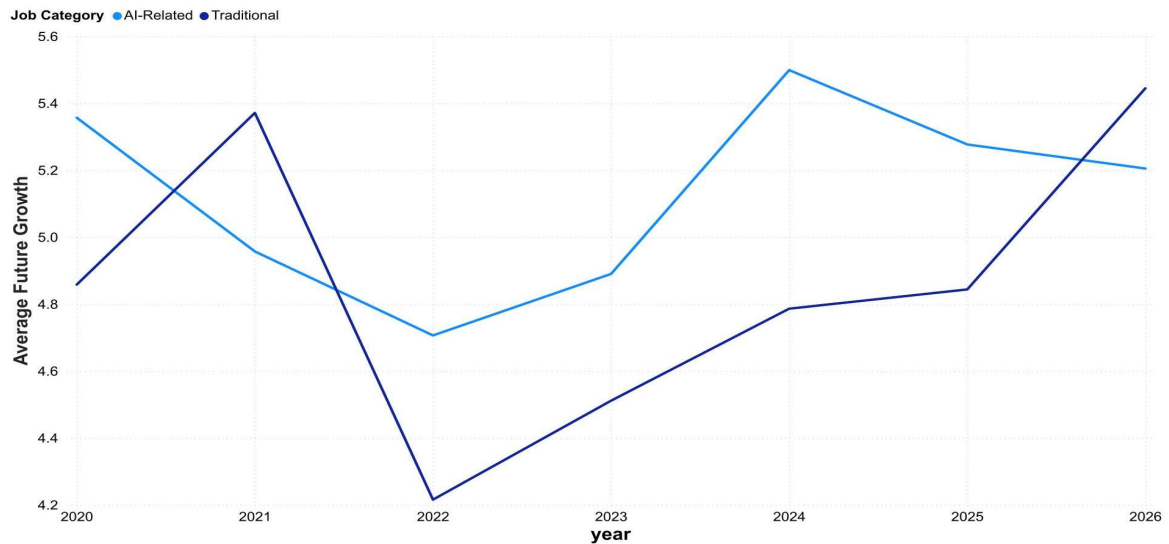
AI Replacement Growth Trend (2020–2026)



The line chart demonstrates a steady increase in AI replacement scores from 2020 to 2026, reflecting the growing influence of automation across industries.

Q19. AI-Related vs Traditional Job Growth

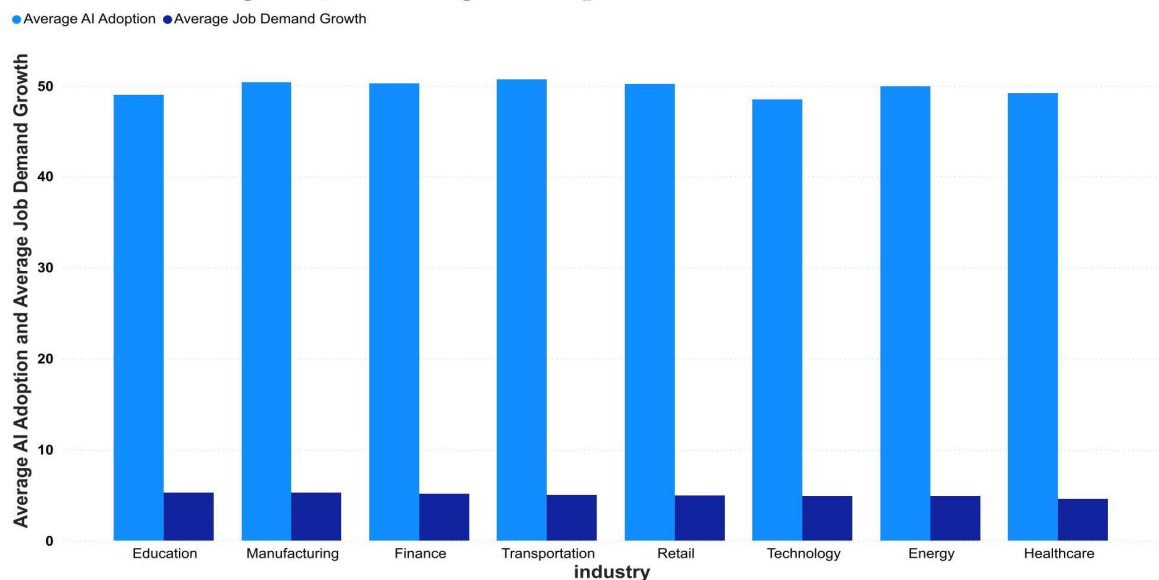
Future Growth Trends: AI-Related vs Traditional Jobs



The comparison shows that AI-related jobs exhibit stronger future growth trends than traditional jobs, indicating an increasing shift toward technology-driven careers.

Q20. Industries Creating New Jobs Through AI

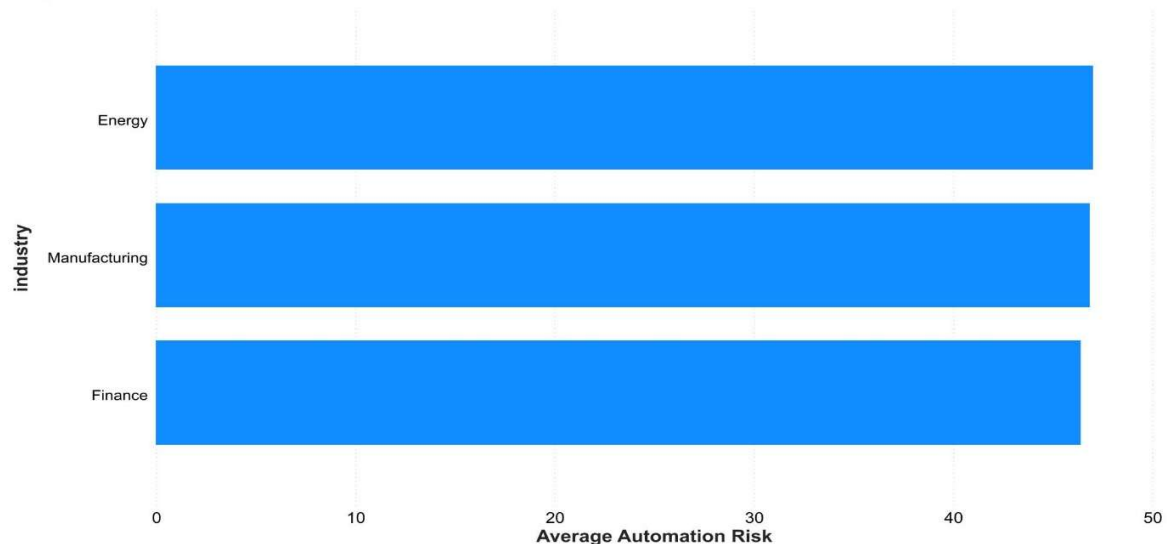
Industries Creating New Jobs Through AI Adoption



The dashboard suggests that industries with higher AI adoption also show stronger future job demand growth, indicating that AI is creating new technology-driven opportunities in several sectors.

Q21. TOP 3 Industries Most Vulnerable to AI Automation

Top 3 Industries Most Vulnerable to AI Automation



Based on the automation risk analysis, Energy, Manufacturing, and Finance appear to be the most vulnerable industries due to their dependence on repetitive and process-oriented tasks.

Q22. Suggest 2 strategies employees can adopt to remain relevant in an AI-driven economy?

Strategy 1:

Employees should continuously upskill themselves in AI, data analysis, and digital technologies to remain competitive in a rapidly evolving workforce.

Strategy 2:

Employees should focus on creativity, communication, leadership, and critical thinking skills because these human-centric abilities are harder for AI systems to automate.

Q23. Which skills should students focus on to reduce automation risk? Justify with data?

Students should focus on AI, programming, data analytics, and digital technology skills because the dashboard shows that technical and high-skill jobs demonstrate stronger future demand growth and comparatively lower automation risk. Additionally, creativity, problem-solving, and leadership skills are important because AI-resistant jobs typically require human judgment and decision-making abilities.

**Q24: Which industry should receive maximum AI investment in the future?
Why?**

The Technology industry should receive maximum AI investment because it demonstrates high AI adoption levels, strong future job demand growth, and major opportunities for innovation and productivity improvement. The analysis indicates that technology-related roles continue to expand despite rising automation trends.

Q25. Which visualization in your dashboard provides the strongest business or societal insight? Explain briefly?

The “Salary vs AI Automation Risk” scatter plot provides the strongest societal insight because it clearly demonstrates the relationship between salary levels and automation risk. The visualization shows that higher-paying and specialized jobs generally experience lower automation risk, while repetitive and lower-skill roles remain more vulnerable to AI disruption.

Strategic Recommendations

- Employees should continuously upskill themselves in AI, analytics, and digital technologies.
- Students should focus on technical, analytical, and problem-solving skills to remain competitive.
- Industries vulnerable to automation should invest in workforce reskilling programs.
- AI investment should prioritize industries where automation enhances productivity rather than completely replacing human workers.

Conclusion

This case study demonstrates that Artificial Intelligence is significantly reshaping industries, workforce structures, and future job opportunities. While AI automation may replace repetitive and predictable tasks, it is also creating new opportunities in technology-driven and analytical fields.

The analysis highlights the growing importance of adaptability, continuous learning, and human-centric skills such as creativity, leadership, and critical thinking. Organizations, employees, and students must prepare for an AI-driven future through reskilling and strategic workforce transformation.

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